Computational and Experimental Aerodynamic Analysis of Nose Cone Geometries

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***Abstract*—Aerodynamic drag plays a critical role in the performance and efficiency of vehicles operating in fluid environments. This study investigates the aerodynamic behavior of three nose cone geometries—flat, rounded, and pointed—through a combination of wind tunnel testing and Computational Fluid Dynamics (CFD) simulations. Airflow velocities were measured using a Pitot tube and digital manometer, while drag forces were quantified using a Linear Variable Differential Transformer (LVDT). CFD models were developed in COMSOL Multiphysics to replicate the experimental conditions and predict aerodynamic performance. Experimental and computational results demonstrated that the flat nose cone generated significantly greater drag than the streamlined geometries. The rounded and pointed nose cones exhibited substantially lower drag coefficients, highlighting the benefits of aerodynamic shaping. While the CFD models successfully captured overall performance trends, discrepancies between simulation and experimental data revealed the impact of mesh resolution, geometric approximation, and testing conditions on simulation accuracy. These findings demonstrate the value of combining computational modeling with physical validation when evaluating aerodynamic designs.**

I. Introduction

The interaction between fluids and solid bodies is fundamental to numerous engineering disciplines, including aerospace, automotive, marine, and biomedical engineering. One of the most significant forces arising from fluid flow is aerodynamic drag, which opposes motion and directly impacts energy consumption, efficiency, and overall system performance. Engineers often optimize the geometry of structures and vehicles to minimize drag and improve operational effectiveness.

Aerodynamic design principles are evident in a wide range of applications, from aircraft fuselages and rocket nose cones to high-performance automobiles. Streamlined geometries reduce flow separation and pressure drag, enabling objects to move more efficiently through fluid environments. Understanding the relationship between geometry and aerodynamic performance is therefore critical in engineering design.

Two complementary approaches are commonly used to investigate aerodynamic behavior: experimental testing and numerical simulation. Wind tunnels provide controlled environments for directly measuring aerodynamic forces and validating theoretical predictions. Computational Fluid Dynamics (CFD), meanwhile, enables engineers to visualize flow behavior, evaluate design alternatives, and predict performance before physical prototypes are constructed.

This investigation compares the aerodynamic performance of flat, rounded, and pointed nose cone geometries using both wind tunnel experimentation and CFD simulations. Airflow velocity and drag force measurements were collected experimentally and compared against numerical predictions generated in COMSOL Multiphysics. Reynolds numbers and drag coefficients were calculated to evaluate aerodynamic performance across multiple flow conditions. By comparing simulation and experimental results, this study assesses the effectiveness of CFD as a predictive design tool while illustrating the influence of geometry on aerodynamic drag.

II. Methods and Analysis

A. Experimental Procedure

Airflow velocity within the wind tunnel was measured using a Pitot tube connected to a digital manometer. The Pitot tube determines dynamic pressure by measuring the difference between total pressure and static pressure, allowing air velocity to be calculated.

Three nose cone geometries were tested during the investigation. The geometries included a flat-faced nose cone, a hemispherical nose cone, and a pointed nose cone. Dimensions of the three models are shown in Fig. 1.

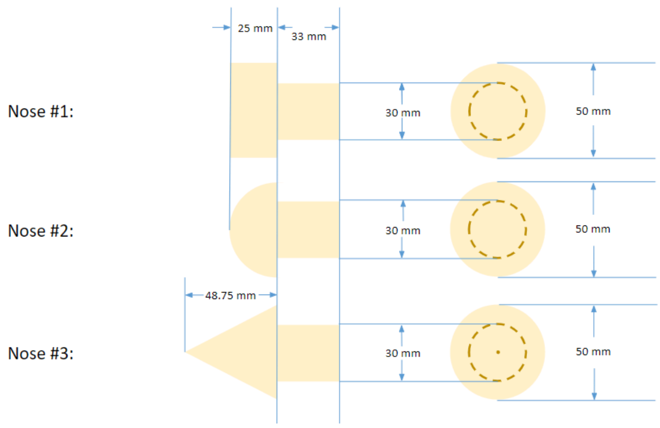

Fig. 1. Dimensions of the flat, rounded, and pointed nose cone geometries. Taken from E80:Lab 6 Lab Manual [2].

Drag forces were measured using a Linear Variable Differential Transformer (LVDT) over a range of airflow velocities. Prior to collecting experimental data, baseline measurements were obtained with the wind tunnel operating at zero airflow to determine sensor offsets. These offsets were subtracted from subsequent measurements to obtain corrected drag force values.

B. Computational Fluid Dynamics Modeling

CFD simulations were performed using COMSOL Multiphysics to replicate the wind tunnel testing conditions. Airflow was modeled using a laminar flow formulation with prescribed inlet velocities corresponding to the experimental measurements. Each nose cone geometry was discretized using a free tetrahedral mesh to approximate the fluid domain and capture aerodynamic behavior.

Simulation results were generated for the same operating conditions used during physical testing, enabling direct comparison between experimental and computational results.

C. Aerodynamic Calculations

To characterize aerodynamic performance, Reynolds numbers and drag coefficients were calculated for each geometry.

The Reynolds number was determined using

$$Re = \frac{\upsilon L}{\nu} \tag{1}$$

where V is the fluid velocity, L is the characteristic length of the nose cone, and ν is the kinematic viscosity of air.

The drag coefficient was calculated using

$$C_d = \frac{2D}{\rho V^2 A} \tag{2}$$

where D is the drag force, ρ is the fluid density, V is the fluid velocity, and A is the reference area.

Reynolds number provides a dimensionless measure of flow behavior by relating inertial forces to viscous forces. The drag coefficient normalizes the drag force with respect to velocity and geometry, enabling direct comparison between different designs.

III. Results and Discussion

A. Wind Tunnel Calibration

The calibration relationship between fan rotational speed and airflow velocity is presented in Fig. 2.

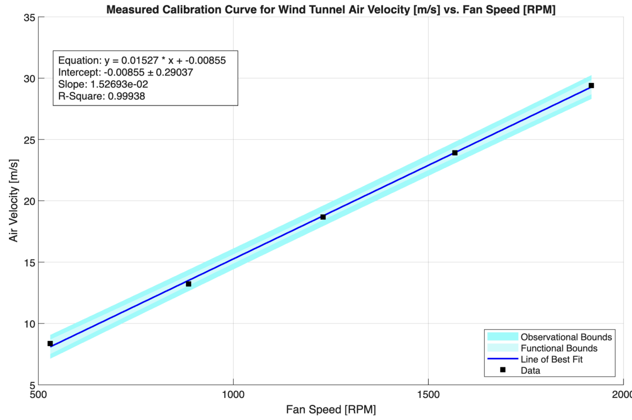

Fig. 2. Wind tunnel calibration curve relating fan RPM to airflow velocity.

As expected, the calibration data demonstrated a strong linear relationship between fan speed and airflow velocity. This relationship enabled airflow conditions to be reproduced consistently throughout testing and simplified subsequent calculations involving Reynolds number and drag coefficient.

B. Drag Force Analysis

Measured and simulated drag force results as a function of Reynolds number are shown in Fig. 3.

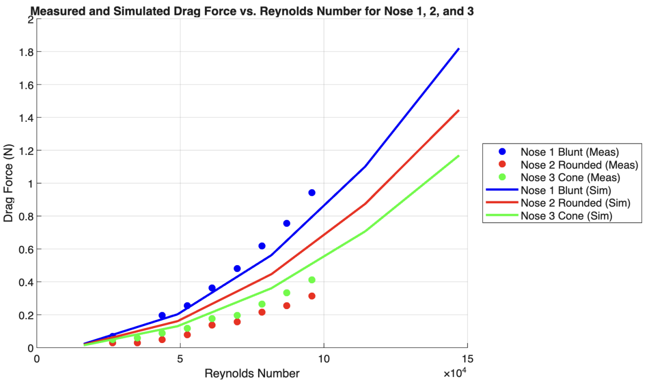

Fig. 3. Experimental and simulated drag force versus Reynolds number for the three nose cone geometries.

For all three geometries, drag force increased with Reynolds number due to increasing airflow velocity. The flat nose cone consistently generated the highest drag force because its geometry creates a large stagnation region and promotes significant pressure drag. In contrast, the rounded and pointed geometries redirected airflow more efficiently, resulting in substantially lower drag forces.

Although CFD simulations captured the overall trends observed experimentally, quantitative differences were present. For the flat geometry, experimentally measured drag coefficients exceeded simulation predictions. One likely explanation is slight angular misalignment during testing, which would increase the effective frontal area exposed to the airflow and therefore increase measured drag.

C. *Drag Coefficient Analysis*

The drag coefficient results are presented in Fig. 4.

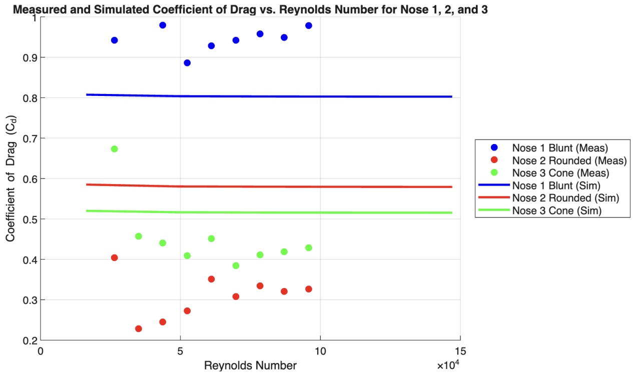

Fig. 4. Experimental and simulated drag coefficient versus Reynolds number for the three nose cone geometries.

The rounded and pointed nose cones exhibited significantly lower drag coefficients than the flat geometry, demonstrating the aerodynamic advantages of streamlined designs. However, CFD simulations predicted higher drag coefficients for the rounded and pointed geometries than were observed experimentally.

The discrepancy was most pronounced for the rounded geometry. Because the CFD model utilized a relatively coarse tetrahedral mesh, smooth curved surfaces were approximated using numerous flat elements. This approximation introduced artificial flow disturbances and increased predicted drag. As a result, the CFD model overestimated the drag coefficient for the rounded nose cone.

The pointed geometry exhibited smaller discrepancies because its profile contains more linear features and is therefore less sensitive to mesh discretization effects. Nevertheless, numerical approximations and limitations in representing complex flow structures likely contributed to the remaining differences between simulation and experimental results.

Overall, the CFD simulations successfully predicted the relative aerodynamic performance of each geometry despite the observed quantitative discrepancies.

IV. Conclusion

This investigation combined wind tunnel testing and Computational Fluid Dynamics simulations to evaluate the aerodynamic performance of flat, rounded, and pointed nose cone geometries. Experimental and computational results demonstrated that geometry strongly influences aerodynamic drag characteristics.

The flat nose cone consistently produced the highest drag forces and drag coefficients due to its blunt frontal surface and increased flow separation. The rounded and pointed geometries generated significantly lower drag, confirming the effectiveness of streamlined designs in reducing aerodynamic resistance.

While CFD simulations accurately captured the general trends observed experimentally, differences between measured and predicted results revealed limitations associated with mesh quality, geometric approximation, and experimental setup variability. In particular, coarse mesh resolution contributed to drag overprediction for curved geometries.

These findings highlight the importance of validating CFD models with experimental measurements and demonstrate how numerical simulations and physical testing can be used together to guide aerodynamic design decisions. Future work should focus on mesh refinement, improved turbulence modeling, and enhanced experimental alignment procedures to further improve agreement between computational and experimental results.

References

[1] RRT Automotive, “Aerodynamic Design,” RRT Automotive, 2026. https://rrtautomotive.com/blogs/racing-services/aerodynamic-design?srsltid=AfmBOoopAUVe8izwAhKLT080qp46uOKmRd65ieA9yzuCGX-yERuHT2wD [Accessed: March 12, 2026]

[2] E80, “Lab 6: Fluid Dynamics,” Harvey Mudd College E80, 2026. https://hmc-e80.github.io/labs/lab6/ [Accessed: March 10, 2026]